

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seven Semester of B. Tech. (CL) Examination

May 2014

CL401 Estimation Costing and Financial concept - I

Date: 20.05.2014(Tuesday)

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Which are the information require to prepare detail estimation of civil works? 04
- (b) Why detail specifications are important for any construction projects? 04
- (c) For road projects, which type of contract generally adopted by government? State the limitation of it. 04
- Q - 2 (a) Distinguish clearly between Approximate estimate and detailed estimate. Also state importance of both for project sanctioning. 05
- (b) What is unit of measurement of plaster work? State the mode of measurement and deduction criteria of plasterwork. 06

OR

- Q - 2 (a) Prepare approximate estimate of estimate of a building having plinth area equal to 2500 sq.m. Given that 05
- | | |
|---|-----------------------|
| i. Plinth area rate | 1600/- per sq.m |
| ii. Architectural works | 2 % of building cost |
| iii. Electrical installation | 10 % of building cost |
| iv. Water supply and sanitary works | 6% of building cost |
| v. Extra for other services | 8% of building cost |
| vi. Contingencies and supervision charges | 10% over total cost |
- (b) State measurement unit of following items. 06
- | | |
|--------------------------------------|--------------------------------|
| a. Pointing on concrete work | b. 70 mm precast concrete jail |
| c. Earth filling in trenches | d. Painting on wood work |
| e. Cement Plastering of 30 cm height | f. Roofing with C.G.I. sheets |

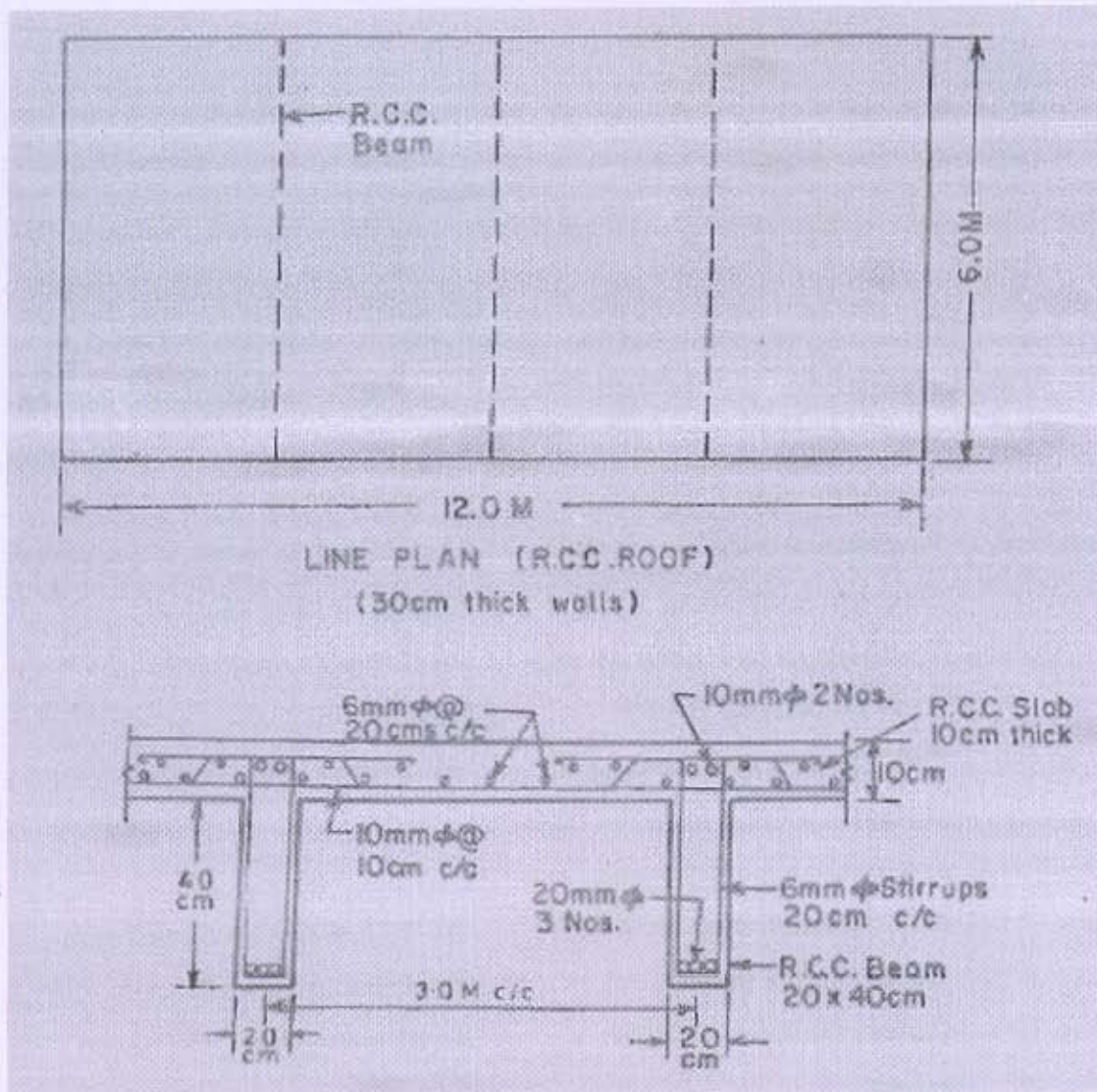
- Q - 3 (a) Write detailed specifications for first class brick work in C.M. (1:6). 06
- (b) What is tender? What do understand by an unbalanced tender? Explain with an example. 06

OR

- Q - 3 (a) Write detailed specifications for Plastering work. 06
- (b) Differentiate between security deposit and retention money. Under which conditions retention money is kept by client? 06

SECTION - II

- Q - 4 (a) Calculate quantity for construction of hall, shown in Figure1, For the following items of work. 13



- i. Calculate the quantity of reinforced concrete work only for beams provided also along the periphery of plan.
- ii. Calculate quantity % steel for beam.
- iii. Calculate quantity of shuttering for beam only

Q - 5 (a) State factors affecting rate of item of work. **04**

(b) For Brick work Above P.L. in CM 1:6, derive following. **08**

- i. Calculate the material required for 5 m^3 of brick work for 230 mm brick wall.
- ii. Calculate the no of days required to finish 5 m^3 of brick work with one mason and its supporting gang with standard task turn out.

OR

Q - 5 (a) Derive rate of 12 cm R.C.C. slab per m^3 having proportion (1:2:4) with 1% steel reinforcement. Assume suitable data if required. **12**

Q - 6 Attempt any **one** **10**

- (a)** Estimate the quantities from the plan of one room shown in Figure 2 (a) & (b) for the following item. Prepare an abstract with relevant market rate.
 - a. Calculate quantity of earth work.
 - b. Calculate quantity of first class brick work using centerline method.
 - c. Calculate quantity of first class flooring work.
- (b)** Estimate the quantities R.C.C and steel from the Figure 3, showing details of retaining wall.

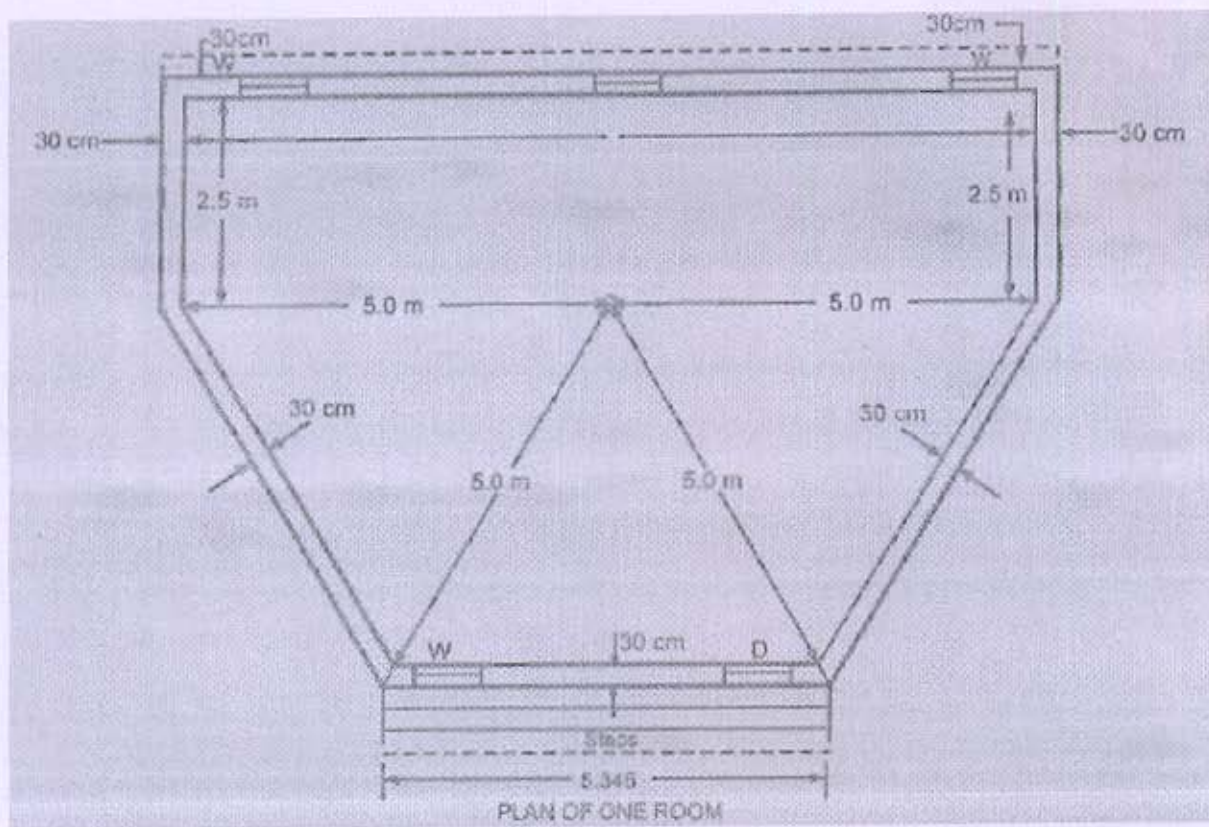


Figure 2 (a) Plan

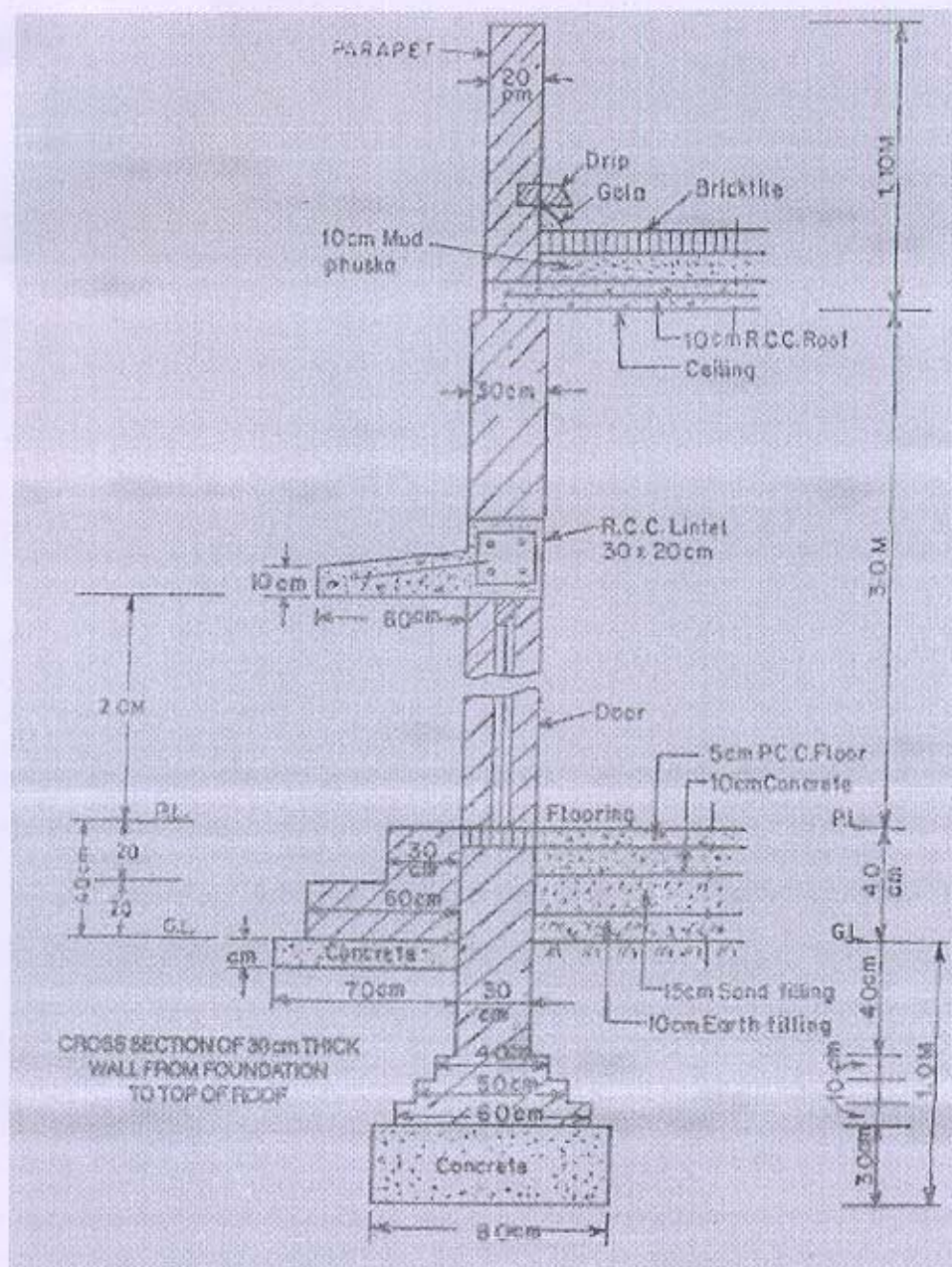


Figure 2 (b) Sectional Elevation

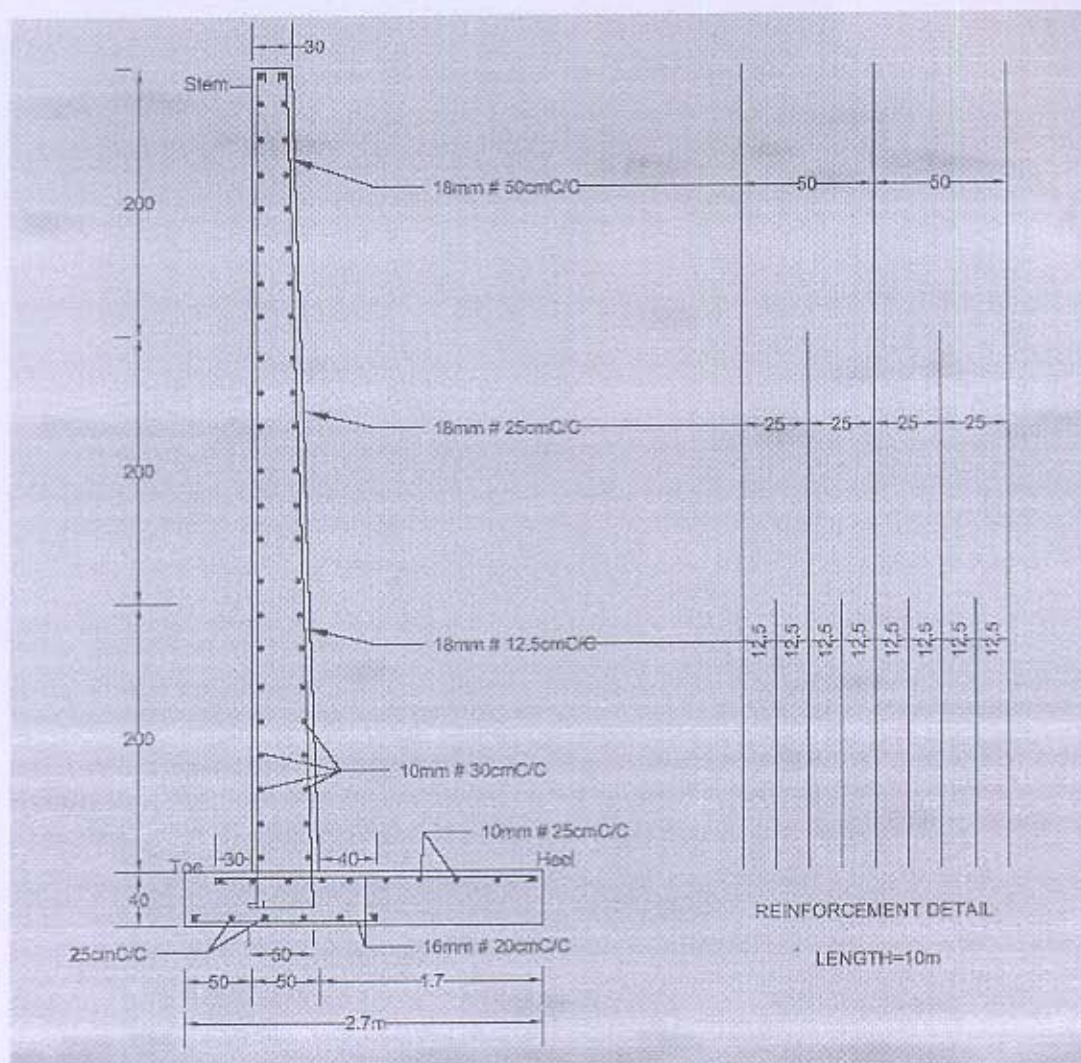


Figure 3
